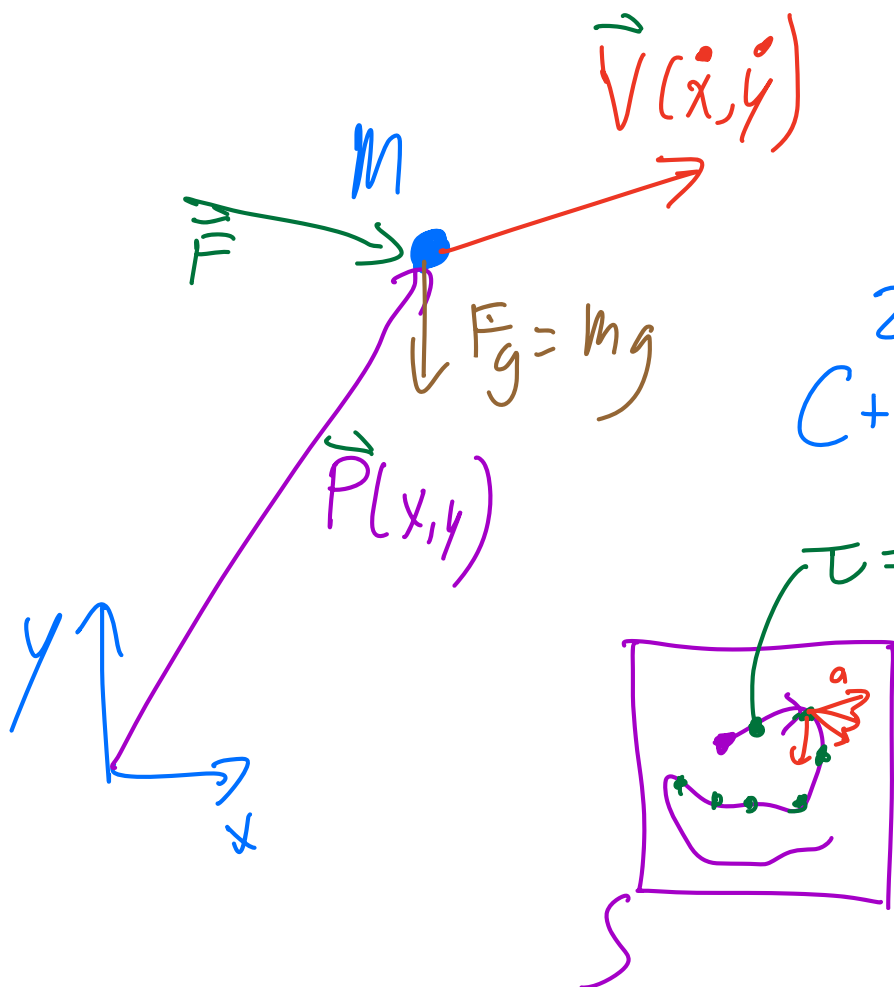


State and Action Spaces for Movement Control

Controllable 2D Point Mass

$$\dot{x} \triangleq \frac{dx}{dt}$$



Configuration

$$C = \begin{bmatrix} x \\ y \end{bmatrix}$$

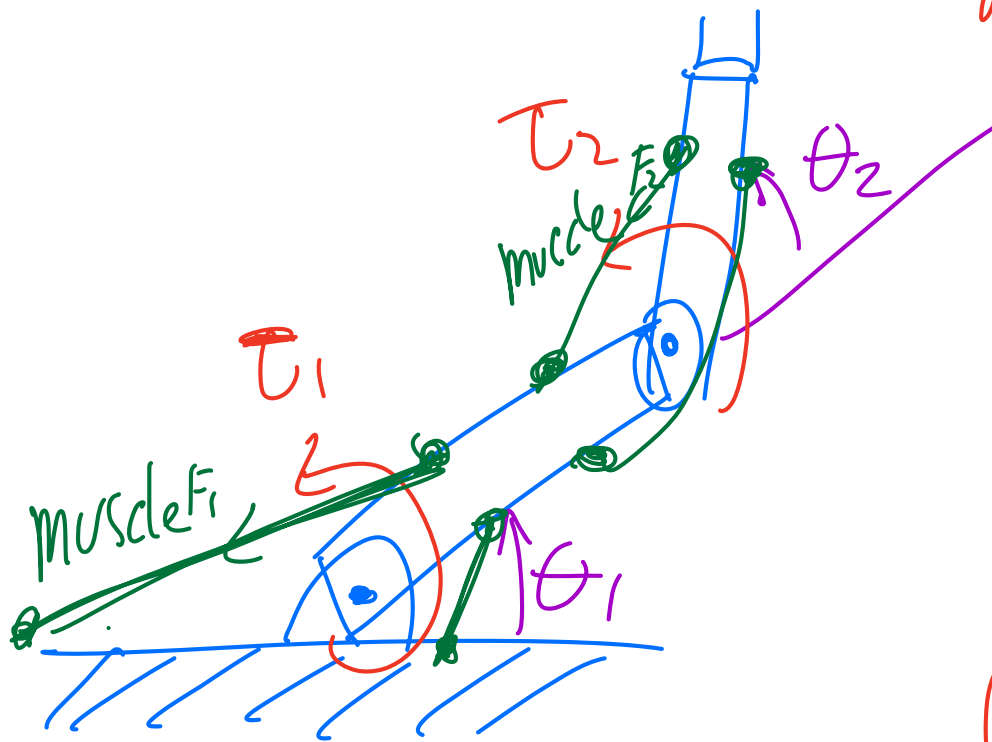
$2 \times$
 $C + \dot{C}$

$$a = \begin{bmatrix} F_x \\ F_y \end{bmatrix} \in \mathbb{R}^2$$

$$S = \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \end{bmatrix} = \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \end{bmatrix} \in \mathbb{R}^4$$

$$3D: S \in \mathbb{R}^6, a \in \mathbb{R}^3$$

Planar 2-link Robot



fully
actuated

$$C = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$$

$$S = \begin{bmatrix} \theta_1 \\ \dot{\theta}_1 \\ \theta_2 \\ \dot{\theta}_2 \end{bmatrix} \text{ or } \begin{bmatrix} \theta_1 \\ \omega_1 \\ \theta_2 \\ \omega_2 \end{bmatrix} \text{ or } \begin{bmatrix} \theta_1 \\ \theta_2 \\ \omega_1 \\ \omega_2 \end{bmatrix}$$

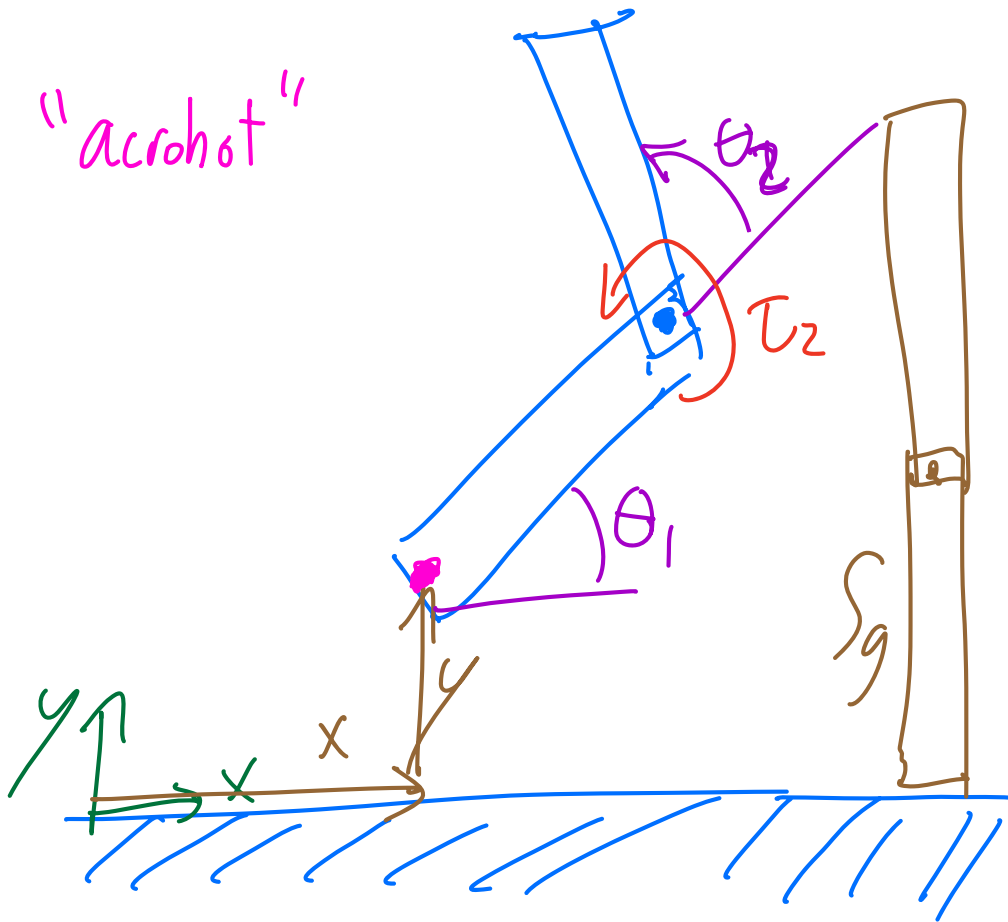
$$d = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix}$$

$$\begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} = f(S, \tau_1, \tau_2)$$

$$\tau_1 = f(\text{muscle}_1)$$

Underactuated Planar 2-link Robot

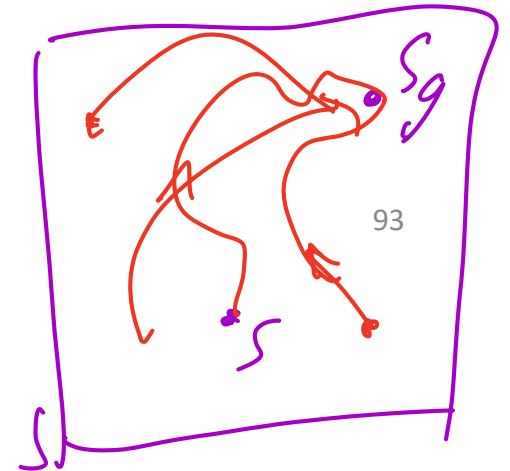
"acrobot"



$C = \begin{bmatrix} x \\ y \\ \theta_1 \\ \theta_2 \end{bmatrix} \in \mathbb{R}^4$ $Q = [\tau_2]$

underactuated

$S = \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ \theta_1 \\ \dot{\theta}_1 \\ \theta_2 \\ \dot{\theta}_2 \end{bmatrix} \in \mathbb{R}^8$



Humanoid Robot / Simple Human Model

Underactuated: $\dim[a] < \dim[C]$

$$C \in \mathbb{R}^{26} + \mathbb{R}^6$$

3D position
3D orientation

$$S \in \mathbb{R}^{52} + \mathbb{R}^{12} = \mathbb{R}^{64}$$

global position &
orientation

$$a \in \mathbb{R}^{26}$$

